

UNIT 3: CSS, Links and Images

CSS Links

- Links can be styled with any CSS property (e.g. color, font-family, background, etc.).
- In addition, links can be styled differently depending on what state they are in.

The four links states are:

a:link - a normal, unvisited link

a:visited - a link the user has visited

a:hover - a link when the user mouses over it

a:active - a link the moment it is clicked

Styling HTML Table

The following are the styles that can be applied for the table:

- Border
- Background color
- Alignment
- Size
- padding
- layout
- hoverable table

Border:

Apply border to table.

```
table{  
border: 1px solid;  
border-collapse:collapse;  
border-spacing:10px;  
}
```

Table Size:

The size of the table can be changed by using the width and height properties.

```
table{  
height:  
width: }
```

table layout:

The table-layout property specifies the structure and behavior of the table.

It can take one of the following values:

- auto: adjusts the column widths automatically based on content
- fixed: specifies the fixed column width

Alignment:

The text-align property sets the horizontal alignment of the table content

The vertical-align property sets the vertical alignment of the table content.

Background Color:

The table can be styled with different background colors.

Table Padding:

The padding property adds the space between the border and the content.

Hoverable Table:

We can add the hover effect to the table elements with the help of the hover pseudo-class selector. For example,

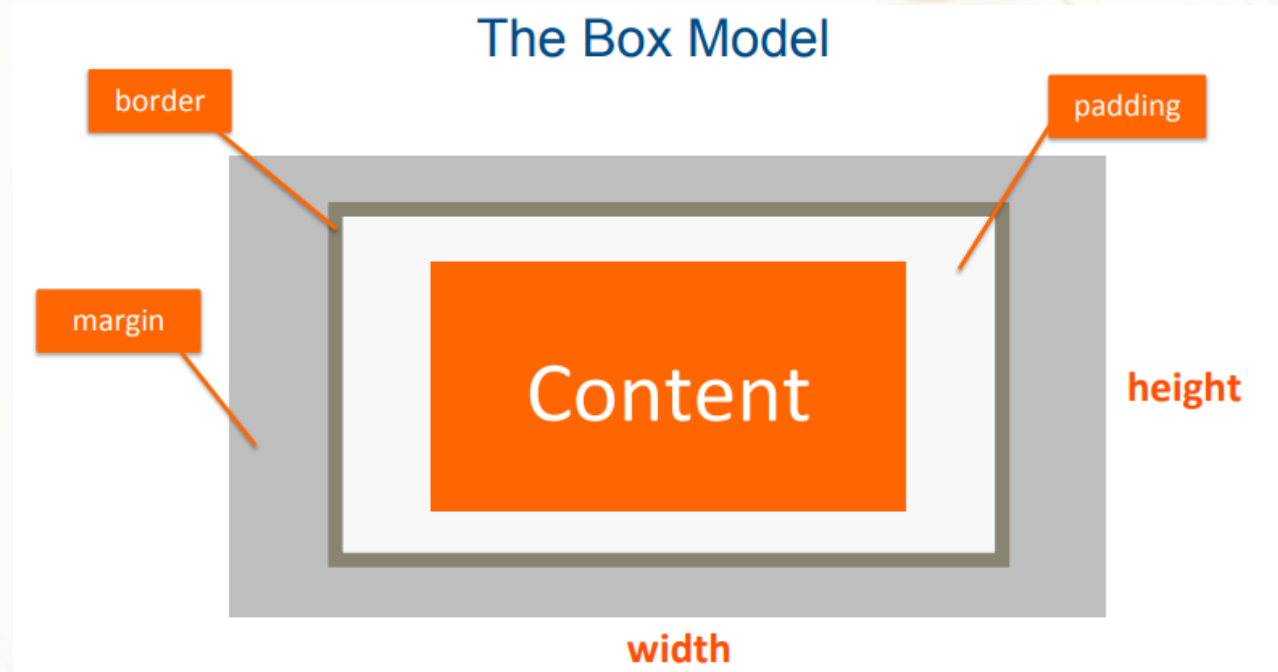
```
tr:hover {  
    background-color: grey;  
}
```


Styling HTML Forms

The following are the styles that can be applied for the input field:

- Width
- Padding
- Margin
- Border
- Border radius
- Box shadow
- Font
- Colored input
- Focused input

The Box Model

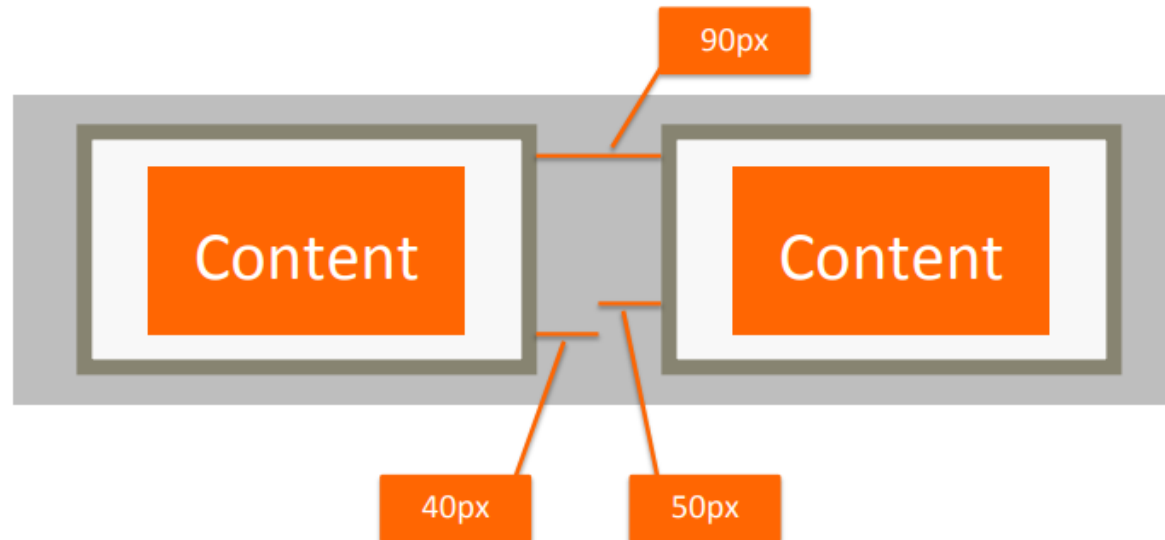


In CSS, the term "box model" is used when talking about design and layout. The CSS box model is essentially a box that wraps around every HTML element. It consists of: margins, borders, padding, and the actual content.

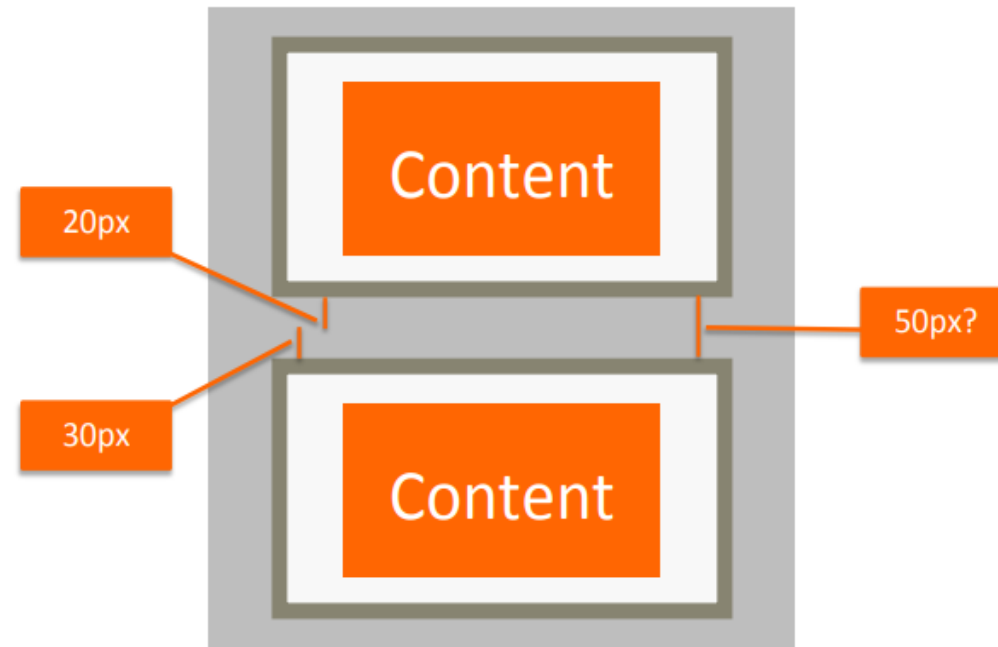
- Content - The content of the box, where text and images appear
- Padding - Clears an area around the content. The padding is transparent
- Border - A border that goes around the padding and content
- Margin - Clears an area outside the border. The margin is transparent

Cumulative Margins

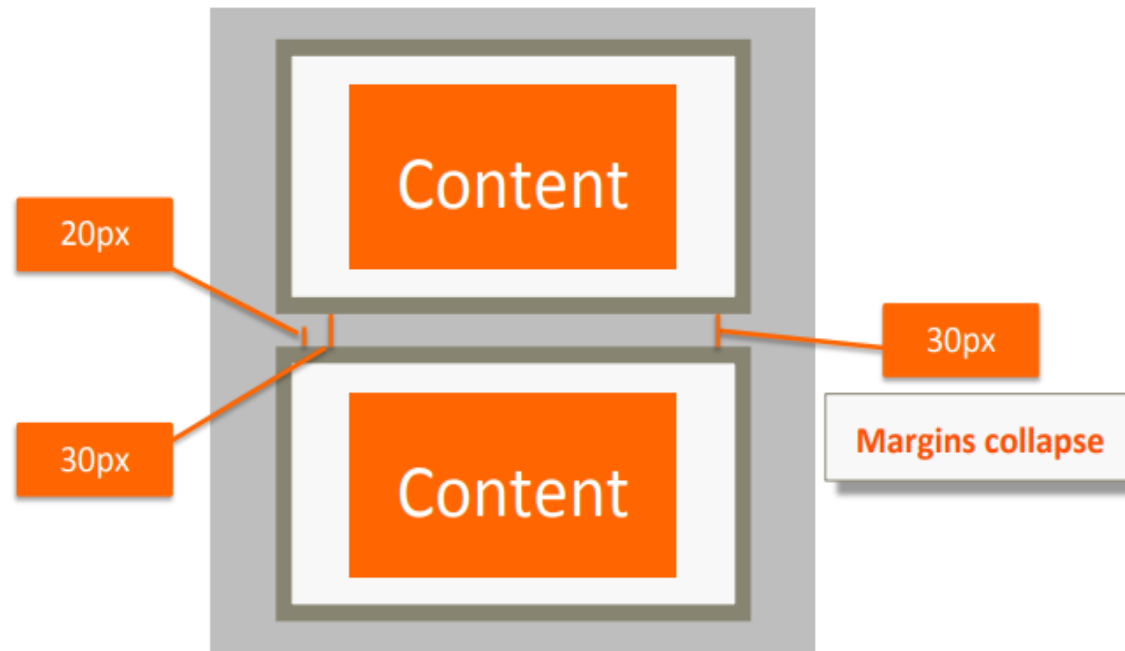
Cumulative Margins



Cumulative Margins?



Cumulative Margins



CSS Styling Images

We can use CSS to style our image in the following ways:

- Changing the image size
- Adding a rounded corner to the image
- Centering image
- Creating responsive image
- Creating thumbnail image
- Creating transparent image
- Adding a background image

CSS Styling Images

Rounded Images

- Use the border-radius property to create rounded images

Thumbnail Images

- Use the border property to create thumbnail images.

Responsive Images

- Responsive images will automatically adjust to fit the size of the screen.
- Resize the browser window to see the effect
- The max-width property is used to make the image responsive.

Center an Image

- To center an image, set left and right margin to auto and make it into a block element.

CSS Styling Images

Transparent Image

- The opacity property can take a value from 0.0 - 1.0. The lower value, the more transparent the image will be.

Changing the Image Size

- The width and height properties are used to resize the image.

CSS Layout – Floating Elements

- Floating elements can produce flexible layout.
- Positioning elements by floating in CSS is a layout technique that allows you to control the alignment of block-level elements within a container.
- When you float an element, you essentially move it to the left or right within its container, and other elements flow around it. This is often used for creating multi-column layouts, wrapping text around images, and more.
- The clear property controls the flow next to floated elements.
- The clear property specifies what should happen with the element that is next to a floating element.

CSS Layout - Position

The position property specifies the type of positioning method used for an element.

There are five different position values:

static

relative

fixed

absolute

sticky

Elements are then positioned using the top, bottom, left, and right properties.

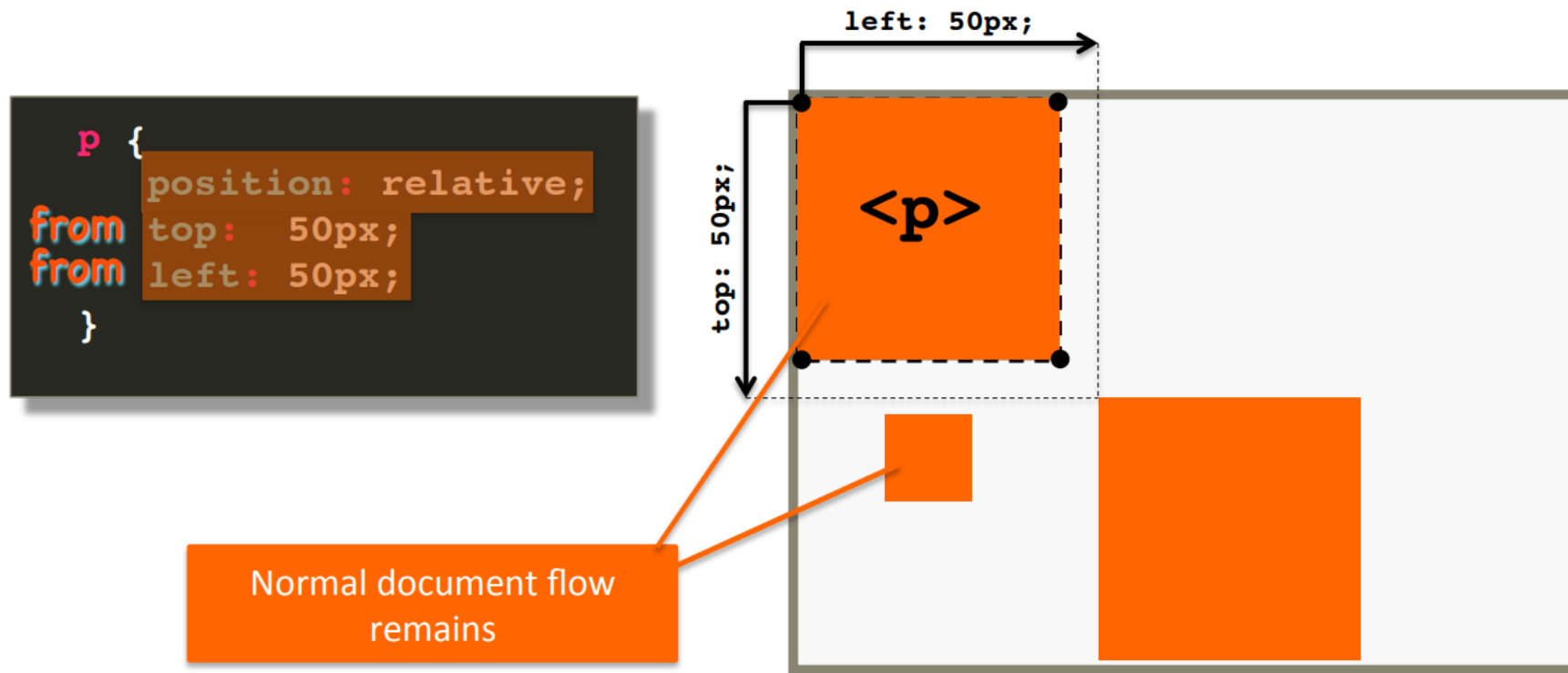
CSS Layout - Position

Static

- HTML elements are positioned static by default.
- Static positioned elements are not affected by the top, bottom, left, and right properties.

Relative Positioning

- Relative positioning is the position where the element is positioned relative to its normal position.



- Absolute positioning

Absolute positioning is the position where the element is positioned absolutely to its first positioned parent. That means an element with position: absolute is removed from the normal document flow. By default, it is positioned to the starting point (top-left corner) of its parent element. We can use the positioning attributes top, left, bottom. and right to set the location.

CSS Layouts



- A header is usually located at the top of the website (or right below a top navigation menu).
- A navigation bar contains a list of links to help visitors navigating through your website:
- Content : The layout in this section, often depends on the target users. The most common layout is one (or combining them) of the following:
 - 1-column (often used for mobile browsers)
 - 2-column (often used for tablets and laptops)
 - 3-column layout (only used for desktops)

Dropdown menu

- HTML) Use any element to open the dropdown content, e.g. a `<span>`, or a `<button>` element.
- Use a container element (like `<div>`) to create the dropdown content and add whatever you want inside of it.
- Wrap a `<div>` element around the elements to position the dropdown content correctly with CSS.
- CSS) The `.dropdown` class uses `position:relative`, which is needed when we want the dropdown content to be placed right below the dropdown button (using `position:absolute`).

CSS Specificity

- CSS specificity is a concept that determines which CSS rule is applied to an HTML element when multiple conflicting styles are defined. It's a way to prioritize and resolve conflicts in styles.
- When multiple CSS rules target the same element, the browser calculates specificity to determine which style to apply. If two or more rules have the same specificity, the one that appears later in the stylesheet takes precedence (i.e., the "cascading" part of CSS).

```
<div id="myElement" class="myClass" style="color: green;">This is  
green text.</div>
```

- In this case, the inline style takes precedence because it has the highest specificity. If you remove the inline style, the ID selector's style (#myElement) will be applied due to its higher specificity, overriding the class selector's style (.myClass).

Origin Precedence (when in conflict)

simple rule

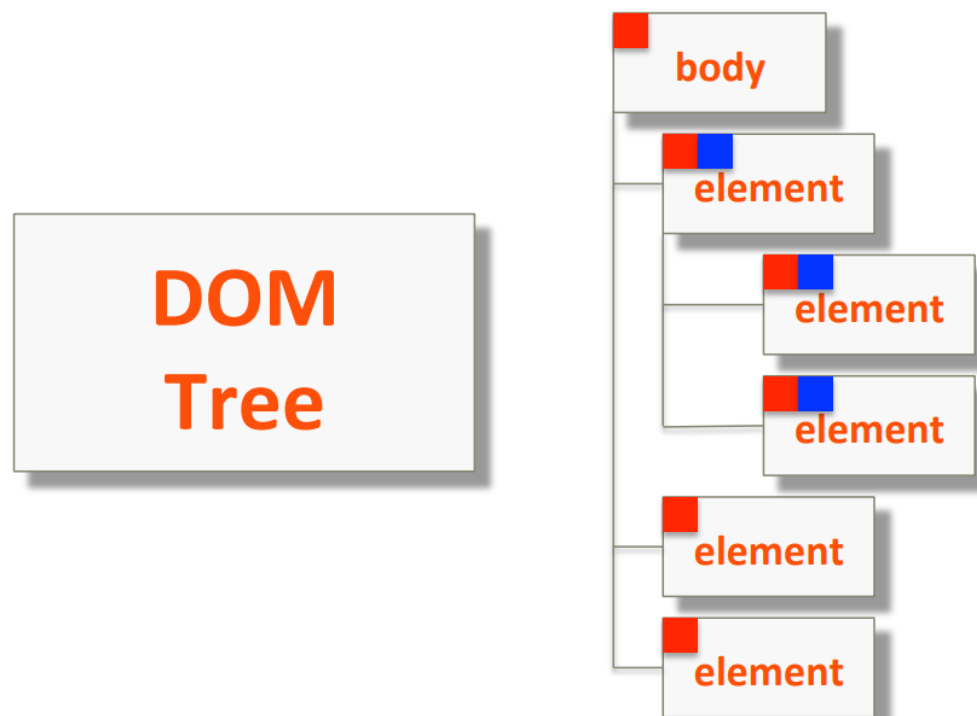
Last Declaration Wins

Origin Precedence (when no conflict)

even simpler rule

Declarations Merge

Inheritance



Specificity

simple rule

**Most Specific
Selector Combination Wins**

CSS Units

- In CSS (Cascading Style Sheets), units are used to define measurements for various properties such as length, width, height, font size, and more. There are several different types of units in CSS, each with its own purpose and usage.

Absolute Units:

- Pixel (px): The pixel is a common unit of measurement in CSS. It's a fixed-size unit and is the same on all devices. One pixel is a single dot on the screen.

Relative Units:

- Percentage (%): Percentage units are relative to the parent element or the container's size. For example, setting the width of an element to 50% will make it half the width of its parent element.

Em (em): The "em" unit is relative to the font size of the parent element. For example, if the font size of a parent element is 16px and you set a child element's font size to 1em, it will be 16px. If you set it to 2em, it will be 32px.

Rem (rem): Similar to the "em" unit, but it's relative to the root (the <html>) element's font size. This makes it easier to maintain consistent sizing throughout your website.